Lossless Decomp → Final qstn

SID	Name	Dept	Dloc
1	a	CSE	B2
2	b	EEE	B1
3	c	CSE	B2

R

→ not a coincidence

→ required rule

True for Entirety of DB

R_1 SID Name Dept

R_2 [Dept Dloc]

$R_1 \cap R_2$ = Dept → R_2

can determine

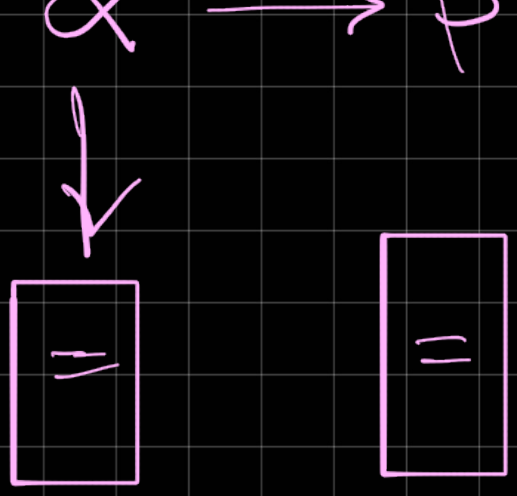
(functional dependency)

$K \subseteq \alpha\beta$ $X = \alpha\gamma\sim$

$t_1 \neq t_2$

ab ab

ac ≠ adr



1st record = alpha

2nd record = β

Dept & Dloc →

$\alpha \rightarrow \beta$ always true.

then dept → dloc

if $\beta \subseteq \alpha$

Dept. Dloc → Dloc.

not requires to check → always true

① → trivial dependency.

② Non-Trivial: No common elements

in between.

$$\alpha \longrightarrow \beta$$

$$\alpha \cap \beta = \emptyset$$

We gotta check. Can't instantly say.

SID, Name ⟶ Hall, CGPA.

(1.5) Semi trivial:

$$\alpha \rightarrow \beta$$

$$\beta \not\subseteq \alpha$$

$$\alpha \cap \beta \neq \emptyset$$

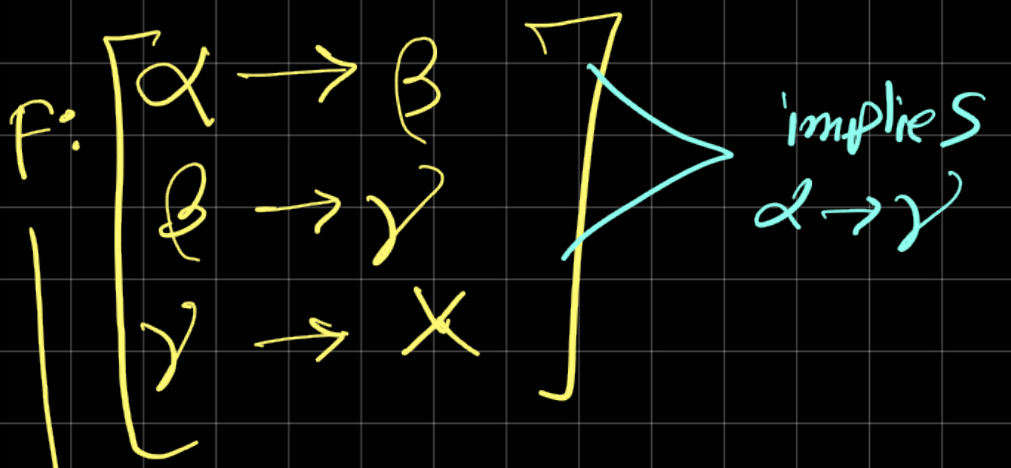
Gotta check

SID, Name ⟶ Hall, Name

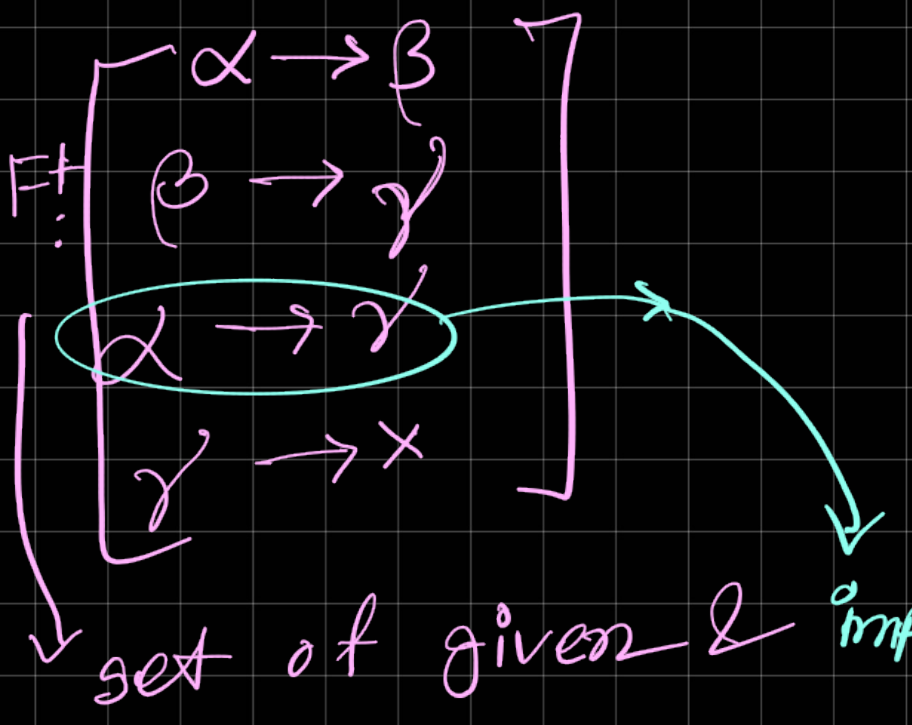
common element exists.

SID → Dept dept → dloc

SID → dloc



set of functional dependency.



set of given & implied func dep.

Augment → both sides → identical stuff add.